

The table shows the number of tons of blueberries that were imported to the US from 2005 to 2013.

Year	Blueberries Imported (tons)
2005	26,319
2006	32,602
2007	35,125
2008	52,111
2009	60,450
2010	76,743
2011	87,336
2012	96,367
2013	103,502

1. Determine the average rate of change from 2005 to 2013.

$$\begin{array}{l} (2005, 26,319) \\ (2013, 103,502) \end{array} \quad \frac{103,502 - 26,319}{2013 - 2005} = \frac{77,183}{8} = 9,647.875$$

2. Interpret the average rate of change for the interval 2005 to 2013 in terms of the context.

From 2005 to 2013, an average of 9,647.875 more tons of blueberries were imported each year.

3. Determine the average rate of change from 2007 to 2010.

$$\begin{array}{l} (2007, 35125) \\ (2010, 76743) \end{array} \quad \frac{76,743 - 35,125}{2010 - 2007} = \frac{41,618}{3} = 13,872.\bar{6}$$

4. Interpret the average rate of change for the interval 2007 to 2010 in terms of the context.

From 2007 to 2010, an average of 13,872. $\bar{6}$ more tons of blueberries were imported each year.

5. How does the average rate of change for the interval 2007 to 2010 compare to the average rate of change for the interval 2005 to 2013?

The average for 2007 – 2010 was larger by 4,224.792 tons.

The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, U.S. passengers traveled on a train for the period of 1970 to 2000, where x is the number of years since 1970.

6. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1970 to 1975?

$$1970 \rightarrow x = 0$$

$$m(0) = 6,179$$

$$1975 \rightarrow x = 5$$

$$m(5) = 5(25) - 160(5) + 6,179$$

$$m(5) = 5504$$

$$\frac{5504-6179}{5-0} = \frac{-675}{5} = -135$$

7. Interpret the average rate of change for the interval 1970 to 1975 in terms of the context.

From 1970 to 1975, the average decreased by 135 million miles yearly.

8. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1975 to 1980?

$$m(5) = 5504$$

$$1975 \rightarrow x = 5$$

$$m(10) = 500 - 1600 + 6179$$

$$1980 \rightarrow x = 10$$

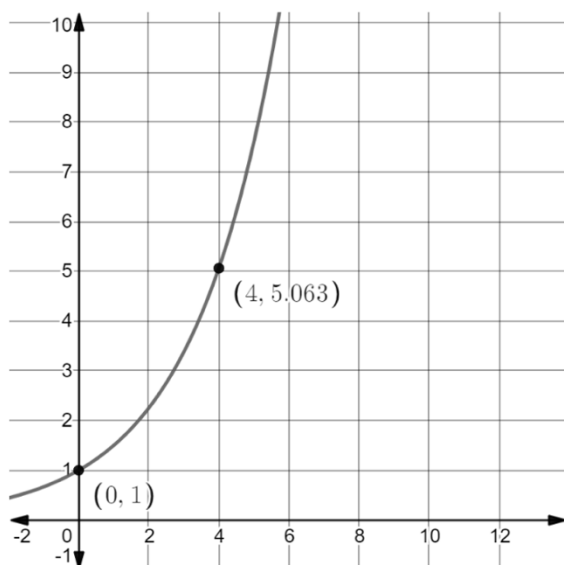
$$m(10) = 5079$$

$$\frac{5079-5504}{10-5} = \frac{-425}{5} = -85$$

9. Interpret the average rate of change for the interval 1975 to 1980 in terms of the context.

From 1975 to 1980, the average decreased by 85 million miles yearly.

10. The graph of a function is shown. Points $(0, 1)$ and $(4, 5.063)$ are on the function. Calculate the average rate of change for the line that passes through the points $(0, 1)$ and $(4, 5.063)$.



$$\frac{5.063-1}{4-0} = \frac{4.063}{4} = 1.01575$$